

5	C	Assume Charlie throws the ball to the right and take rightwards as positive. Consider Charlie and the ball as 1 system, Before throwing the ball, $p_{\text{before throw}} = p_{\text{after throw}}$ $0 = m_{\text{ball}} u_{\text{ball}} + m_{\text{Charlie}} (-u_{\text{Charlie}})$ $0 = (1.5)(2.5) + (60)(u_{\text{Charlie}})$ $u_{\text{Charlie}} = -0.0625 \text{ m s}^{-1}$ Charlie is moving to the left with speed 0.0625 m s^{-1} after throwing the ball. Consider the returning ball and Charlie as 1 system now, $p_{\text{before catch}} = p_{\text{after catch}}$ $(60)(-0.0626) + (1.5)(-2.5) = (61.5)(v_{\text{after catch}})$ $v_{\text{after catch}} = -0.122 \text{ m s}^{-1}$
6	C	$T \cos 18^\circ = W$